Polybrominated diphenyl ethers in U.S. Meat and poultry from two statistically designed surveys showing trends and levels from 2002 to 2008.

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Polybrominated diphenyl ether (PBDE) body burdens in the general U.S. population have been linked to the consumption of red meat and poultry. Exposure estimates have also indicated that meat products are a major contributor to PBDE dietary intake. To establish solid estimates of PBDE concentrations in domestic meat and poultry, samples from two statistically designed surveys of U.S. meat and poultry were analyzed for PBDEs. The two surveys were conducted in 2002-2003 and 2007-2008, between which times the manufacturing of penta-BDE and octa-BDE formulations had ceased in the United States (December 2004). Thus, the data provided an opportunity to observe prevalence and concentration trends that may have occurred during this time frame and to compare the mean PBDE levels among the meat and poultry industries. On the basis of composite samples, the average sum of the seven most prevalent PBDEs (BDE-28, -47, -99, -100, -153, -154, and -183) decreased by >60% from 1.95 ng/g lipid in 2002-2003 to 0.72 ng/g lipid in 2007-2008 for meat and poultry. PBDEs measured in individual samples in 2008 showed that beef samples had the lowest PBDE levels followed by hogs and chickens and then by turkeys. The PBDE congener pattern was the same for both surveys and resembled the penta-BDE formulation with BDE-47 and -99 accounting for 30 and 40% of the total, respectively. On the basis of the data from the two surveys, it appears that PBDE levels in U.S. meat and poultry have declined since manufacturing ceased; however, exposure pathways of PBDEs to livestock are still not known.